

Networking Planning document review

Planning Document Review: Networking Team (Group 2: Kilo)

Timing Issues and Missed Milestones

- **Vague Phase Deadlines:** Each phase outlines objectives but lacks clear start and end dates. This could lead to confusion and missed milestones.
- **Sequential Dependencies:** Phase 2 relies on the successful completion of Phase 1 without overlap. Delays in Phase 1 (e.g., StubPlayer implementation) will follow to subsequent phases.

Risks Impacting Completion

- **Move Validation Complexity:** Improving move validation (Phase 2) is described as an enhancement but is critical to game integrity. If not prioritized, it could result in invalid game states.
- **StubPlayer Reliability:** Randomized move generation needs to handle edge cases (e.g., invalid moves or repeated turns). Without robust logic, testing outcomes will be unreliable.
- **Lack of AI Integration Plan:** While AI is noted as a future possibility, no framework is planned for easy incorporation if required later.

Suggestions for Improvement

1. **Parallel Development:** Allow Phase 2 tasks (move validation and StubPlayer refinement) to begin during Phase 1 to avoid delays.
2. **Validation Priority:** Move validation should be treated as a core deliverable in Phase 1 rather than deferred to later stages.
3. **AI-Ready Structure:** Implement interfaces or hooks in GameSessionHandler to allow future AI integration without major rewrites.
4. **Edge Case Testing:** Design targeted tests for StubPlayer to verify move validity under various scenarios (e.g., turn tracking, win conditions).

General Recommendations:

- **Cross-Team Communication:** Ensure regular updates between groups to align stub implementation with actual networking design.
- **Documentation Updates:** Regularly update planning documents to reflect changes, especially if milestones or tasks shift.
- **Proactive Risk Management:** Identify and address risks early (e.g., incomplete testing, unclear dependencies) through regular reviews and feedback cycles.